Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Caton Farms

Physical address of dairy:

15635 S Steinegul RD	Escalon	San Joaquin	95320
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X229-X090-X050-XXXX

B. OPERATORS

Caton, Cody

Operator name: Caton, Cody

Telephone no.: (209) 456-3266 (Landline) (Cellular)

15206 Steinegul RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Barbara Beckett Bypass Trust

Legal owner name: Barbara Beckett Bypass Trust

Telephone no.: (509) 679-0665 (Landline) (Cellular)

1840 North RD	Wenatchee	WA	98801
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	100	105	100	100	86
Number under roof	475	0	0	0	0	0
Maximum number	475	100	105	100	100	86
Average number	475	100	105	100	100	86
Avg live weight (lbs)	1,400	1,450	900	650		

Predominant milk cow breed: Holstein

Average milk production: 69 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,227.90 *tons per reporting period*

Total nitrogen from manure: 205,785.78 *lbs per reporting period*

After ammonia losses (30% loss applied): 144,050.05 *lbs per reporting period*

Total phosphorus from manure: 33,208.86 *lbs per reporting period*

Total potassium from manure: 90,654.81 *lbs per reporting period*

Total salt from manure: 246,648.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 18,861,270 *gallons*

Total nitrogen generated: 38,538.81 *lbs*

Total phosphorus generated: 10,900.12 *lbs*

Total potassium generated: 139,530.45 *lbs*

Total salt generated: 619,510.60 *lbs*

	18,861,270	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	18,861,270	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/29/2023	Solid commercial fertilizer *4-10-10*	11.80 *ton*	Dry-weight	0.1	4.000000	10.000000	10.000000	0.000000
06/20/2023	Solid commercial fertilizer *UN32*	37.04 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	24,624.95	2,357.64	2,357.64	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	24,624.95	2,357.64	2,357.64	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	18	18	2	process wastewater	X229-X090-XX05-XXXX
10	27	27	2	process wastewater	X229-X090-XX05-XXXX
11	16	16	2	manure	X229-X090-XX05-XXXX
12	8	8	2	manure	X229-X090-XX05-XXXX
13	27	27	2	manure	X229-X090-XX05-XXXX
2	16	16	2	both	X229-X090-XX05-XXXX
3	30	30	2	process wastewater	X229-X090-XX05-XXXX
4	37	37	2	manure	X229-X090-XX05-XXXX
5	25	25	2	process wastewater	X229-X090-XX05-XXXX
6	36	36	1	manure	X229-X090-XX05-XXXX
7	26	26	2	manure	X229-X080-XX80-XXXX X229-X080-XX81-XXXX X229-X080-XX82-XXXX
8	15	15	0	none	X229-X080-XX79-XXXX X229-X080-XX80-XXXX X229-X080-XX81-XXXX X229-X080-XX82-XXXX
9	19	19	2	process wastewater	X229-X090-XX05-XXXX
P3	13	13	2	manure	X229-X090-XX05-XXXX
Totals for areas that were used for application	298	298	25		
Totals for areas that were not used for application	15	15	0		
Land application area totals	313	313	25		

B. CROPS AND HARVESTS

1

Field name: 1

1

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	240.45 *ton*	As-is		68.8	4,200.00	1,000.00	14,000.00		8.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.36	112.21	26.72	374.03	726.03

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	506.60 *ton*	As-is		63.2	3,500.00	500.00	2,600.00		4.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.14	197.01	28.14	146.35	890.72

10

Field name: 10

10/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	356.40 *ton*	As-is		68.2	4,000.00	900.00	13,400.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.20	105.60	23.76	353.76	720.31

10

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	697.70 *ton*	As-is		66.2	3,200.00	500.00	2,300.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.84	165.38	25.84	118.87	768.61

11

Field name: 11

10/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 10/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	216.00 *ton*	As-is		65.4	4,500.00	600.00	3,500.00		5.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.50	121.50	16.20	94.50	536.23

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	420.80 *ton*	As-is		66.6	3,200.00	400.00	2,400.00		2.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.30	168.32	21.04	126.24	509.48

12

Field name: 12

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	102.00 *ton*	As-is		65.4	4,500.00	600.00	3,500.00		5.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.75	114.75	15.30	89.25	506.44

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	223.20 *ton*	As-is		64.6	3,000.00	700.00	3,500.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.90	167.40	39.06	195.30	1,106.18

13

Field name: 13

10/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 10/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	369.90 *ton*	As-is		68.2	4,000.00	900.00	13,400.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.70	109.60	24.66	367.16	747.59

13

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	707.40 *ton*	As-is		69.2	3,600.00	400.00	3,900.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.20	188.64	20.96	204.36	677.85

2

Field name: 2

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	225.60 *ton*	As-is		69.4	4,400.00	800.00	11,300.00		2.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.10	124.08	22.56	318.66	233.85

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	400.00 *ton*	As-is		63.2	3,500.00	500.00	2,600.00		4.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.00	175.00	25.00	130.00	791.20

3

Field name: 3

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	393.00 *ton*	As-is		65.4	4,500.00	600.00	3,500.00		5.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.10	117.90	15.72	91.70	520.34

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	777.00 *ton*	As-is		69.4	3,400.00	500.00	3,900.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.90	176.12	25.90	202.02	808.39

4

Field name: 4

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	503.20 *ton*	As-is		74.2	4,000.00	500.00	3,200.00		5.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.60	108.80	13.60	87.04	394.39

4

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	973.10 *ton*	As-is		68.6	3,200.00	600.00	3,000.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.30	168.32	31.56	157.80	660.66

5

Field name: 5

10/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	322.50 *ton*	As-is		74.2	4,000.00	500.00	3,200.00		5.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.90	103.20	12.90	82.56	374.09

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	645.00 *ton*	As-is		69.2	3,600.00	400.00	3,900.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.80	185.76	20.64	201.24	667.50

6

Field name: 6

09/30/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 36 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	340.00 *ton*	As-is		11.8	24,700.00	2,300.00	21,500.00		9.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.44	466.56	43.44	406.11	1,589.36

7

Field name: 7

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	381.40 *ton*	As-is		74.2	4,000.00	500.00	3,200.00		5.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.67	117.35	14.67	93.88	425.40

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	657.80 *ton*	As-is		69.8	3,200.00	700.00	4,700.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.30	161.92	35.42	237.82	687.65

9

Field name: 9

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	243.20 *ton*	As-is		68.2	4,000.00	900.00	13,400.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.80	102.40	23.04	343.04	698.48

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	494.00 *ton*	As-is		69.8	3,200.00	700.00	4,700.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	166.40	36.40	244.40	706.68

P3

Field name: P3

10/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	166.35 *ton*	As-is		69.4	4,400.00	800.00	11,300.00		2.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.80	112.61	20.47	289.19	212.23

P3

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	337.40 *ton*	As-is		69.2	3,600.00	400.00	3,900.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.95	186.87	20.76	202.44	671.48

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/08/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.06	4.21	56.70	596.30	225,650.00 *gal*
Canal	Surface water	1.41	0.00	0.00	72.82	3,020,400.00 *gal*
Application event totals		34.47	4.21	56.70	669.11	

12/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.24	3.08	41.57	437.22	165,450.00 *gal*
Application event totals		24.24	3.08	41.57	437.22	

01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.80	12.36	151.52	286.30	200,500.00 *gal*
Application event totals		19.80	12.36	151.52	286.30	

02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.34	10.82	132.66	250.68	175,555.00 *gal*
Application event totals		17.34	10.82	132.66	250.68	

03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.89	9.30	113.96	215.33	150,800.00 *gal*
Application event totals		14.89	9.30	113.96	215.33	

1 - 10/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.42	10.94	67.78	614.25	380,725.00 *gal*
Canal	Surface water	1.11	0.00	0.00	56.94	2,361,870.00 *gal*
Application event totals		47.53	10.94	67.78	671.19	

1 - 06/13/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.41	0.00	0.00	72.46	3,005,802.00 *gal*
Application event totals		1.41	0.00	0.00	72.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.20	0.00	0.00	61.70	2,559,366.00 *gal*
Application event totals		89.20	0.00	0.00	61.70	

1 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.36	2,337,750.00 *gal*
Application event totals		1.09	0.00	0.00	56.36	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.53	8.44	129.26	1,136.59	600,885.00 *gal*
Canal	Surface water	1.04	0.00	0.00	53.41	2,215,566.00 *gal*
Application event totals		69.57	8.44	129.26	1,190.01	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.62	12.02	184.13	1,619.06	855,950.00 *gal*
Canal	Surface water	1.01	0.00	0.00	52.02	2,157,750.00 *gal*
Application event totals		98.63	12.02	184.13	1,671.08	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.97	0.00	0.00	49.90	2,069,766.00 *gal*
Application event totals		0.97	0.00	0.00	49.90	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.01	0.00	0.00	52.17	2,164,176.00 *gal*
Application event totals		1.01	0.00	0.00	52.17	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	48.73	2,021,473.00 *gal*
Application event totals		0.95	0.00	0.00	48.73	

10 - 10/07/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 10/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.41	0.00	0.00	72.60	4,517,089.00 *gal*
Application event totals		1.41	0.00	0.00	72.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.71	31.04	380.44	718.86	755,150.00 *gal*
Application event totals		49.71	31.04	380.44	718.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.72	32.92	403.42	762.30	800,775.00 *gal*
Application event totals		52.72	32.92	403.42	762.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.20	25.72	315.26	595.71	625,775.00 *gal*
Application event totals		41.20	25.72	315.26	595.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	61.19	3,807,413.00 *gal*
Application event totals		1.19	0.00	0.00	61.19	

10 - 06/12/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/12/2023

10 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.39	0.00	0.00	71.74	4,463,802.00 *gal*
Application event totals		1.39	0.00	0.00	71.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
WWS	Process wastewater	58.99	7.27	111.27	978.43	775,900.00 *gal*
Canal	Surface water	1.27	0.00	0.00	65.61	4,082,454.00 *gal*
Application event totals		148.27	7.27	111.27	1,044.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.97	4.92	75.39	662.88	525,665.00 *gal*
Canal	Surface water	1.18	0.00	0.00	60.88	3,788,246.00 *gal*
Application event totals		41.15	4.92	75.39	723.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.06	0.00	0.00	54.70	3,403,620.00 *gal*
Application event totals		1.06	0.00	0.00	54.70	

10 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.37	3,507,213.00 *gal*
Application event totals		1.09	0.00	0.00	56.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.62	5.62	86.05	756.61	600,000.00 *gal*
Canal	Surface water	1.01	0.00	0.00	52.16	3,245,454.00 *gal*
Application event totals		46.63	5.62	86.05	808.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.04	0.00	0.00	53.79	3,346,612.00 *gal*
Application event totals		1.04	0.00	0.00	53.79	

11 - 10/04/2022: Oats, silage-soft dough

Field name: 11

Crop: Oats, silage-soft dough

Plant date: 10/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	75.79	2,794,620.00 *gal*
Application event totals		1.47	0.00	0.00	75.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	104.00 *ton*
Application event totals		144.27	6.90	46.94	0.00	

11 - 10/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.07	0.00	0.00	54.95	2,026,173.00 *gal*
Application event totals		1.07	0.00	0.00	54.95	

11 - 05/29/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	160.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	75.96	2,800,613.00 *gal*
Application event totals		1.48	0.00	0.00	75.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.21	0.00	0.00	62.22	2,294,030.00 *gal*
Application event totals		89.21	0.00	0.00	62.22	

11 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.11	0.00	0.00	57.25	2,110,887.00 *gal*
Application event totals		1.11	0.00	0.00	57.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.04	0.00	0.00	53.63	1,977,431.00 *gal*
Application event totals		1.04	0.00	0.00	53.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	50.45	1,860,043.00 *gal*
Application event totals		0.98	0.00	0.00	50.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.96	0.00	0.00	49.67	1,831,304.00 *gal*
Application event totals		0.96	0.00	0.00	49.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	48.04	1,771,460.00 *gal*
Application event totals		0.93	0.00	0.00	48.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	48.57	1,790,841.00 *gal*
Application event totals		0.94	0.00	0.00	48.57	

12 - 10/05/2022: Oats, silage-soft dough

12 - 10/05/2022: Oats, silage-soft dough

Field name: 12

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	52.00 *ton*
Application event totals		144.27	6.90	46.94	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	76.38	1,408,071.00 *gal*
Application event totals		1.48	0.00	0.00	76.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.06	0.00	0.00	54.34	1,001,723.00 *gal*
Application event totals		1.06	0.00	0.00	54.34	

12 - 05/29/2023: Corn, silage

Field name: 12

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	80.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

12 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	73.30	1,351,272.00 *gal*
Application event totals		1.42	0.00	0.00	73.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.27	0.00	0.00	65.21	1,202,216.00 *gal*
Application event totals		89.27	0.00	0.00	65.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.13	0.00	0.00	58.13	1,071,672.00 *gal*
Application event totals		1.13	0.00	0.00	58.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	54.29	1,000,920.00 *gal*
Application event totals		1.05	0.00	0.00	54.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	51.68	952,696.00 *gal*
Application event totals		1.00	0.00	0.00	51.68	

12 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.96	0.00	0.00	49.33	909,463.00 *gal*
Application event totals		0.96	0.00	0.00	49.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	48.24	889,413.00 *gal*
Application event totals		0.94	0.00	0.00	48.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	45.06	830,721.00 *gal*
Application event totals		0.88	0.00	0.00	45.06	

13 - 10/04/2022: Oats, silage-soft dough

Field name: 13

Crop: Oats, silage-soft dough

Plant date: 10/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	175.50 *ton*
Application event totals		144.27	6.90	46.94	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.44	0.00	0.00	74.38	4,628,130.00 *gal*
Application event totals		1.44	0.00	0.00	74.38	

13 - 10/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.02	0.00	0.00	52.42	3,261,603.00 *gal*
Application event totals		1.02	0.00	0.00	52.42	

13 - 06/19/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	270.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	73.33	4,562,703.00 *gal*
Application event totals		1.42	0.00	0.00	73.33	

06/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.27	0.00	0.00	65.64	4,083,885.00 *gal*
Application event totals		89.27	0.00	0.00	65.64	

13 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	56.32	3,503,995.00 *gal*
Application event totals		1.09	0.00	0.00	56.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	50.76	3,158,565.00 *gal*
Application event totals		0.99	0.00	0.00	50.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.96	0.00	0.00	49.62	3,087,385.00 *gal*
Application event totals		0.96	0.00	0.00	49.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.85	0.00	0.00	43.71	2,719,726.00 *gal*
Application event totals		0.85	0.00	0.00	43.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.74	0.00	0.00	38.03	2,366,413.00 *gal*
Application event totals		0.74	0.00	0.00	38.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.70	0.00	0.00	36.02	2,241,360.00 *gal*
Application event totals		0.70	0.00	0.00	36.02	

13 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.65	0.00	0.00	33.50	2,084,213.00 *gal*
Application event totals		0.65	0.00	0.00	33.50	

2 - 10/08/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	75.96	2,800,721.00 *gal*
Application event totals		1.48	0.00	0.00	75.96	

01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.29	29.53	361.87	683.77	425,650.00 *gal*
Application event totals		47.29	29.53	361.87	683.77	

02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.82	26.74	327.69	619.20	385,455.00 *gal*
Application event totals		42.82	26.74	327.69	619.20	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.35	23.95	293.49	554.57	345,225.00 *gal*
Application event totals		38.35	23.95	293.49	554.57	

2 - 10/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.08	8.27	51.23	464.24	255,775.00 *gal*
Canal	Surface water	1.00	0.00	0.00	51.70	1,906,423.00 *gal*
Application event totals		36.09	8.27	51.23	515.95	

2 - 06/13/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	160.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.40	2,743,280.00 *gal*
Application event totals		1.45	0.00	0.00	74.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.25	0.00	0.00	64.27	2,369,840.00 *gal*
Application event totals		89.25	0.00	0.00	64.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.12	0.00	0.00	57.72	2,128,048.00 *gal*
Application event totals		1.12	0.00	0.00	57.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	53.27	1,964,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	51.23	1,888,800.00 *gal*
Application event totals		0.99	0.00	0.00	51.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.96	0.00	0.00	49.58	1,828,080.00 *gal*
Application event totals		0.96	0.00	0.00	49.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	47.81	1,762,910.00 *gal*
Application event totals		0.93	0.00	0.00	47.81	

2 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	49.05	1,808,421.00 *gal*
Application event totals		0.95	0.00	0.00	49.05	

3 - 10/01/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.93	6.10	82.21	864.53	545,255.00 *gal*
Canal	Surface water	1.40	0.00	0.00	72.14	4,987,080.00 *gal*
Application event totals		49.33	6.10	82.21	936.67	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.92	2.92	39.32	413.51	260,800.00 *gal*
Application event totals		22.92	2.92	39.32	413.51	

12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.27	3.98	53.64	564.06	355,750.00 *gal*
Application event totals		31.27	3.98	53.64	564.06	

02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.84	19.26	236.00	445.94	520,505.00 *gal*
Application event totals		30.84	19.26	236.00	445.94	

3 - 10/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.74	14.83	181.70	343.34	400,750.00 *gal*
Application event totals		23.74	14.83	181.70	343.34	

3 - 06/14/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.44	0.00	0.00	74.31	5,137,650.00 *gal*
Application event totals		1.44	0.00	0.00	74.31	

06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

06/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	65.20	4,507,350.00 *gal*
Application event totals		1.27	0.00	0.00	65.20	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.11	0.00	0.00	57.13	3,949,680.00 *gal*
Application event totals		89.11	0.00	0.00	57.13	

3 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.76	0.00	0.00	39.21	2,710,970.00 *gal*
Application event totals		0.76	0.00	0.00	39.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	51.06	3,529,680.00 *gal*
Application event totals		0.99	0.00	0.00	51.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.25	5.94	91.00	800.21	705,077.00 *gal*
Canal	Surface water	0.64	0.00	0.00	32.99	2,280,714.00 *gal*
Application event totals		48.89	5.94	91.00	833.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.46	6.95	106.50	936.48	825,150.00 *gal*
Canal	Surface water	0.82	0.00	0.00	42.05	2,907,213.00 *gal*
Application event totals		57.28	6.95	106.50	978.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.06	6.41	98.19	863.39	760,750.00 *gal*
Canal	Surface water	0.85	0.00	0.00	43.84	3,031,170.00 *gal*
Application event totals		52.91	6.41	98.19	907.24	

3 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	39.43	2,726,113.00 *gal*
Application event totals		0.77	0.00	0.00	39.43	

4 - 10/01/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.37	0.00	0.00	70.45	6,007,213.00 *gal*
Application event totals		1.37	0.00	0.00	70.45	

09/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	240.50 *ton*
Application event totals		144.27	6.90	46.94	0.00	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.97	0.00	0.00	49.80	4,246,613.00 *gal*
Application event totals		0.97	0.00	0.00	49.80	

4 - 06/01/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/01/2023

4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	370.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.38	5,915,486.00 *gal*
Application event totals		1.35	0.00	0.00	69.38	

06/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	62.00	5,286,819.00 *gal*
Application event totals		1.20	0.00	0.00	62.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.14	0.00	0.00	58.63	4,999,366.00 *gal*
Application event totals		89.14	0.00	0.00	58.63	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	53.85	4,591,256.00 *gal*
Application event totals		1.05	0.00	0.00	53.85	

4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	51.49	4,390,272.00 *gal*
Application event totals		1.00	0.00	0.00	51.49	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	47.39	4,041,130.00 *gal*
Application event totals		0.92	0.00	0.00	47.39	

07/31/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	45.40	3,871,013.00 *gal*
Application event totals		0.88	0.00	0.00	45.40	

08/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.89	0.00	0.00	45.75	3,900,721.00 *gal*
Application event totals		0.89	0.00	0.00	45.75	

5 - 10/02/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 10/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	70.14	4,041,130.00 *gal*
Application event totals		1.36	0.00	0.00	70.14	

5 - 10/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.93	5.72	77.06	810.44	425,950.00 *gal*
Application event totals		44.93	5.72	77.06	810.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.82	4.81	64.87	682.21	358,554.00 *gal*
Application event totals		37.82	4.81	64.87	682.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.52	24.67	302.40	571.41	555,787.00 *gal*
Application event totals		39.52	24.67	302.40	571.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.45	17.76	217.68	411.32	400,077.00 *gal*
Application event totals		28.45	17.76	217.68	411.32	

5 - 06/14/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	69.14	3,983,150.00 *gal*
Application event totals		1.34	0.00	0.00	69.14	

5 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	61.09	3,519,375.00 *gal*
Application event totals		1.19	0.00	0.00	61.09	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
WWS	Process wastewater	37.42	4.61	70.59	620.69	455,750.00 *gal*
Canal	Surface water	1.09	0.00	0.00	55.87	3,218,875.00 *gal*
Application event totals		126.51	4.61	70.59	676.56	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.53	5.12	78.34	688.85	505,800.00 *gal*
Canal	Surface water	1.03	0.00	0.00	52.96	3,051,250.00 *gal*
Application event totals		42.56	5.12	78.34	741.81	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	50.56	2,912,625.00 *gal*
Application event totals		0.98	0.00	0.00	50.56	

5 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.26	8.65	132.52	1,165.28	855,625.00 *gal*
Canal	Surface water	0.94	0.00	0.00	48.16	2,774,725.00 *gal*
Application event totals		71.19	8.65	132.52	1,213.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.85	0.00	0.00	43.69	2,516,894.00 *gal*
Application event totals		0.85	0.00	0.00	43.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.84	0.00	0.00	43.21	2,489,170.00 *gal*
Application event totals		0.84	0.00	0.00	43.21	

6 - 09/30/2022: Alfalfa, hay

Field name: 6

Crop: Alfalfa, hay

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	221.96	10.62	72.22	0.00	360.00 *ton*
Application event totals		221.96	10.62	72.22	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	65.60	5,442,444.00 *gal*
Application event totals		1.27	0.00	0.00	65.60	

6 - 09/30/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	51.13	4,241,600.00 *gal*
Application event totals		0.99	0.00	0.00	51.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.02	0.00	0.00	52.53	4,357,800.00 *gal*
Application event totals		1.02	0.00	0.00	52.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	44.74	3,711,500.00 *gal*
Application event totals		0.87	0.00	0.00	44.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	50.99	4,230,432.00 *gal*
Application event totals		0.99	0.00	0.00	50.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.80	0.00	0.00	41.27	3,423,500.00 *gal*
Application event totals		0.80	0.00	0.00	41.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	47.21	3,916,620.00 *gal*
Application event totals		0.92	0.00	0.00	47.21	

6 - 09/30/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	50.50	4,189,216.00 *gal*
Application event totals		0.98	0.00	0.00	50.50	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.01	0.00	0.00	51.93	4,308,417.00 *gal*
Application event totals		1.01	0.00	0.00	51.93	

7 - 10/03/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.50	0.00	0.00	77.05	4,616,703.00 *gal*
Application event totals		1.50	0.00	0.00	77.05	

09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	169.00 *ton*
Application event totals		144.27	6.90	46.94	0.00	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	64.95	3,891,261.00 *gal*
Application event totals		1.26	0.00	0.00	64.95	

7 - 05/30/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	260.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	75.57	4,528,134.00 *gal*
Application event totals		1.47	0.00	0.00	75.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.30	0.00	0.00	66.81	4,002,934.00 *gal*
Application event totals		89.30	0.00	0.00	66.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.11	0.00	0.00	57.17	3,425,526.00 *gal*
Application event totals		1.11	0.00	0.00	57.17	

7 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	52.98	3,174,288.00 *gal*
Application event totals		1.03	0.00	0.00	52.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.99	0.00	0.00	51.06	3,059,056.00 *gal*
Application event totals		0.99	0.00	0.00	51.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	48.67	2,916,273.00 *gal*
Application event totals		0.95	0.00	0.00	48.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.23	0.00	0.00	63.57	3,808,723.00 *gal*
Application event totals		1.23	0.00	0.00	63.57	

9 - 10/03/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.37	6.28	84.69	890.62	355,750.00 *gal*
Canal	Surface water	1.47	0.00	0.00	75.51	3,306,171.00 *gal*
Application event totals		50.84	6.28	84.69	966.13	

9 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.27	25.15	308.20	582.37	430,500.00 *gal*
Application event totals		40.27	25.15	308.20	582.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.13	30.68	375.97	710.41	525,155.00 *gal*
Application event totals		49.13	30.68	375.97	710.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.16	0.00	0.00	59.76	2,616,721.00 *gal*
Application event totals		1.16	0.00	0.00	59.76	

9 - 05/29/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.45	0.00	0.00	74.66	3,268,950.00 *gal*
Application event totals		1.45	0.00	0.00	74.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

9 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.08	8.26	126.53	1,112.59	620,875.00 *gal*
Canal	Surface water	1.26	0.00	0.00	65.05	2,848,062.00 *gal*
Application event totals		68.35	8.26	126.53	1,177.64	

06/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
WWS	Process wastewater	48.18	5.93	90.87	799.04	445,900.00 *gal*
Canal	Surface water	1.10	0.00	0.00	56.51	2,474,275.00 *gal*
Application event totals		137.28	5.93	90.87	855.55	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	54.22	2,374,221.00 *gal*
Application event totals		1.05	0.00	0.00	54.22	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.48	4.00	61.27	538.71	300,625.00 *gal*
Canal	Surface water	0.98	0.00	0.00	50.71	2,220,302.00 *gal*
Application event totals		33.47	4.00	61.27	589.42	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.97	0.00	0.00	49.97	2,187,916.00 *gal*
Application event totals		0.97	0.00	0.00	49.97	

9 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.85	0.00	0.00	43.61	1,909,431.00 *gal*
Application event totals		0.85	0.00	0.00	43.61	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.91	0.00	0.00	47.09	2,061,781.00 *gal*
Application event totals		0.91	0.00	0.00	47.09	

P3 - 10/04/2022: Oats, silage-soft dough

Field name: P3

Crop: Oats, silage-soft dough

Plant date: 10/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.27	6.90	46.94	0.00	84.50 *ton*
Application event totals		144.27	6.90	46.94	0.00	

09/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	69.12	2,070,743.00 *gal*
Application event totals		1.34	0.00	0.00	69.12	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	55.65	1,667,130.00 *gal*
Application event totals		1.08	0.00	0.00	55.65	

P3 - 06/19/2023: Corn, silage

Field name: P3

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.54	25.80	82.56	4,403.20	130.00 *ton*
Application event totals		119.54	25.80	82.56	4,403.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.35	0.00	0.00	69.53	2,082,925.00 *gal*
Application event totals		1.35	0.00	0.00	69.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	6.00	15.00	15.00	0.00	
Application event totals		6.00	15.00	15.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	88.00	0.00	0.00	0.00	
Canal	Surface water	1.26	0.00	0.00	64.88	1,943,695.00 *gal*
Application event totals		89.26	0.00	0.00	64.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.13	0.00	0.00	57.96	1,736,345.00 *gal*
Application event totals		1.13	0.00	0.00	57.96	

P3 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.04	0.00	0.00	53.41	1,600,157.00 *gal*
Application event totals		1.04	0.00	0.00	53.41	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.01	0.00	0.00	52.01	1,558,167.00 *gal*
Application event totals		1.01	0.00	0.00	52.01	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.85	0.00	0.00	43.68	1,308,426.00 *gal*
Application event totals		0.85	0.00	0.00	43.68	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.54	0.00	0.00	27.59	826,416.00 *gal*
Application event totals		0.54	0.00	0.00	27.59	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.65	0.00	0.00	33.67	1,008,721.00 *gal*
Application event totals		0.65	0.00	0.00	33.67	

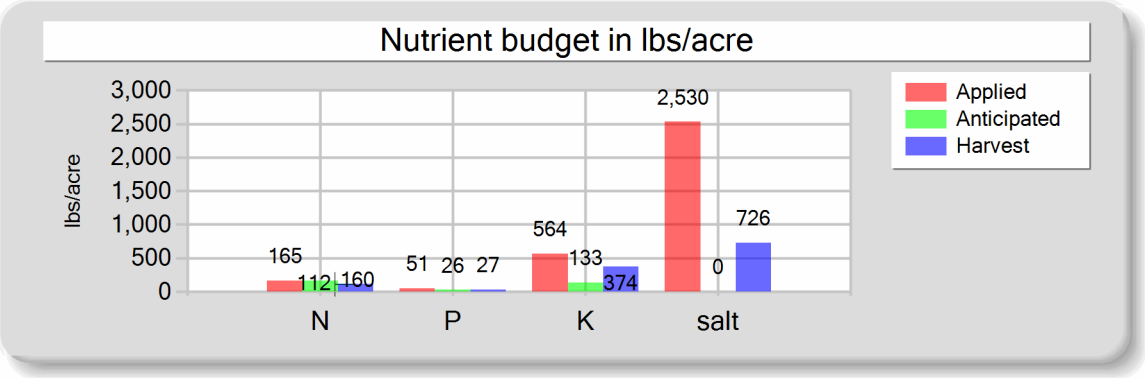
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.60	0.00	0.00	30.73	920,607.00 *gal*
Application event totals		0.60	0.00	0.00	30.73	

B. NUTRIENT BUDGET

1 - 10/08/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	155.74	50.72	564.19	2,400.07
Fresh water	2.52	0.00	0.00	129.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.27	50.72	564.19	2,529.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.21	26.72	374.03	726.03
Nutrient balance	53.06	24.00	190.16	1,803.80
Applied to removed ratio	1.47	1.90	1.51	3.48

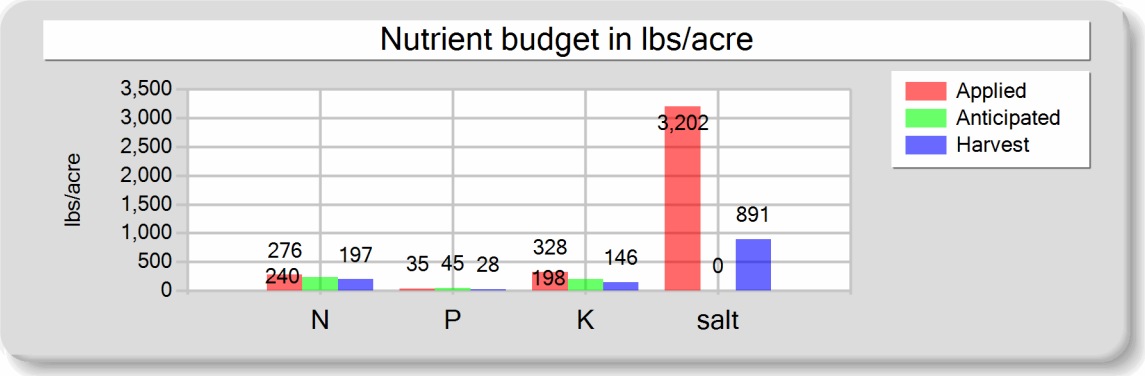
Fresh water applied	
5,382,270.00	*gallons*
198.21	*acre-inches*
11.01	*inches/acre*

Process wastewater applied	
1,298,680.00	*gallons*
47.83	*acre-inches*
2.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/13/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.15	20.46	313.39	2,755.65
Fresh water	8.68	0.00	0.00	446.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.83	35.46	328.39	3,202.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	197.01	28.14	146.35	890.72
Nutrient balance	78.82	7.32	182.04	2,311.69
Applied to removed ratio	1.40	1.26	2.24	3.60

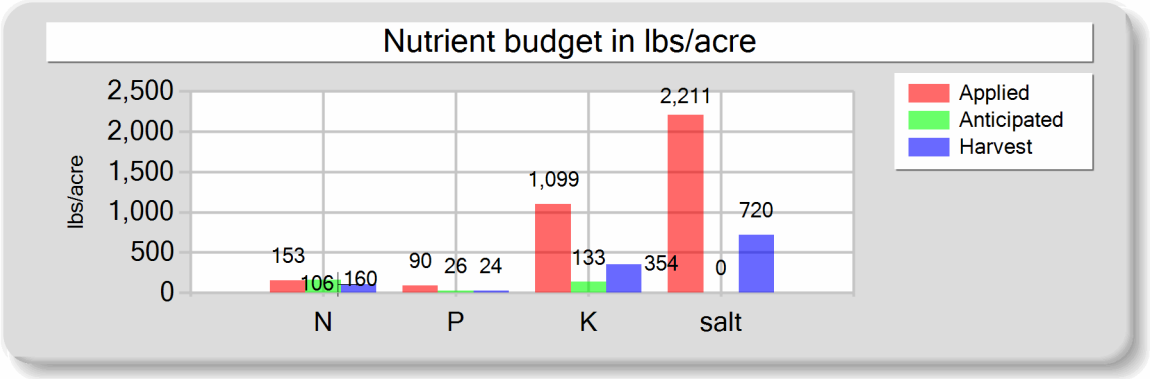
Fresh water applied	
18,531,649.00	*gallons*
682.46	*acre-inches*
37.91	*inches/acre*

Process wastewater applied	
1,456,835.00	*gallons*
53.65	*acre-inches*
2.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 10/07/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 10/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	143.63	89.68	1,099.12	2,076.87
Fresh water	2.60	0.00	0.00	133.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.23	89.68	1,099.12	2,210.66
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	105.60	23.76	353.76	720.31
Nutrient balance	47.63	65.92	745.36	1,490.35
Applied to removed ratio	1.45	3.77	3.11	3.07

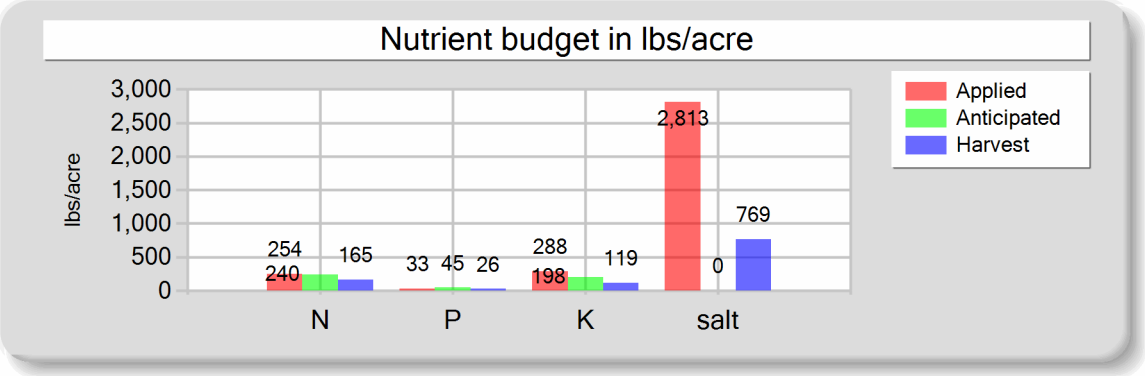
Fresh water applied	
8,324,502.00	*gallons*
306.56	*acre-inches*
11.35	*inches/acre*

Process wastewater applied	
2,181,700.00	*gallons*
80.34	*acre-inches*
2.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/12/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	144.58	17.81	272.70	2,397.92
Fresh water	8.07	0.00	0.00	415.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.65	32.81	287.70	2,813.17
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	165.38	25.84	118.87	768.61
Nutrient balance	88.26	6.97	168.84	2,044.56
Applied to removed ratio	1.53	1.27	2.42	3.66

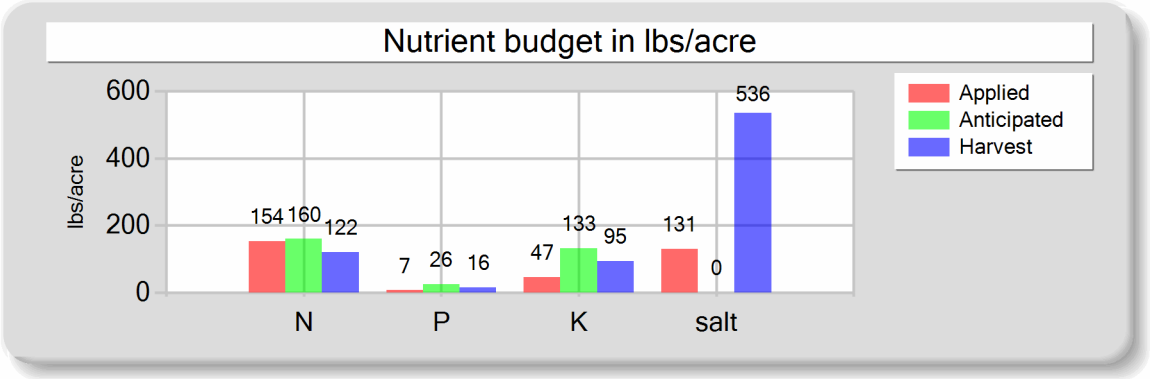
Fresh water applied	
25,837,401.00	*gallons*
951.50	*acre-inches*
35.24	*inches/acre*

Process wastewater applied	
1,901,565.00	*gallons*
70.03	*acre-inches*
2.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 10/04/2022: Oats, silage-soft dough

Field name: 11 Crop: Oats, silage-soft dough Plant date: 10/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.54	0.00	0.00	130.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.81	6.90	46.94	130.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.50	16.20	94.50	536.23
Nutrient balance	32.31	-9.30	-47.56	-405.48
Applied to removed ratio	1.27	0.43	0.50	0.24

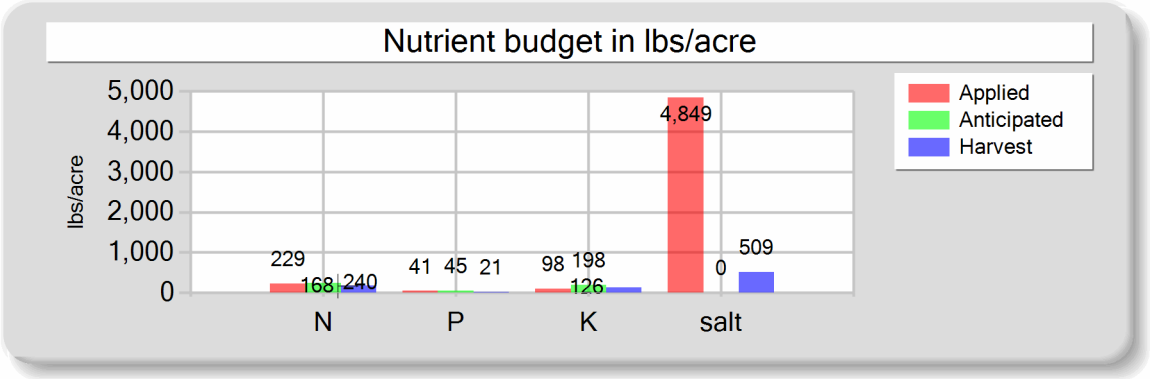
Fresh water applied	
4,820,793.00	*gallons*
177.53	*acre-inches*
11.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 05/29/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.66	0.00	0.00	445.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.20	40.80	97.56	4,848.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	168.32	21.04	126.24	509.48
Nutrient balance	60.88	19.76	-28.68	4,339.50
Applied to removed ratio	1.36	1.94	0.77	9.52

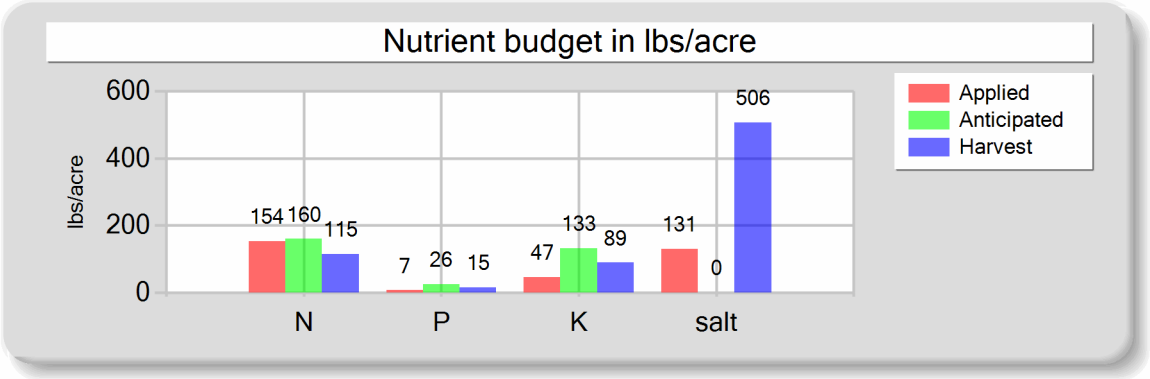
Fresh water applied	
16,436,609.00	*gallons*
605.30	*acre-inches*
37.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 10/05/2022: Oats, silage-soft dough

Field name: 12 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.54	0.00	0.00	130.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.81	6.90	46.94	130.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	114.75	15.30	89.25	506.44
Nutrient balance	39.06	-8.40	-42.31	-375.73
Applied to removed ratio	1.34	0.45	0.53	0.26

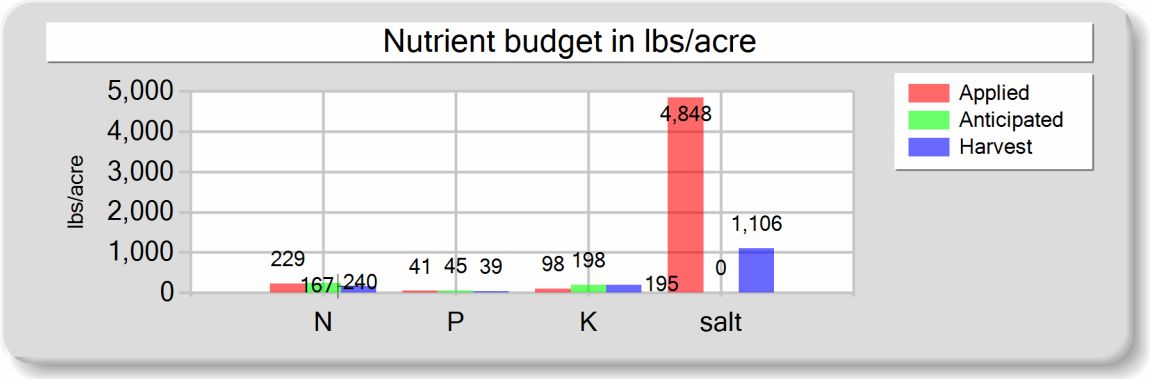
Fresh water applied	
2,409,794.00	*gallons*
88.74	*acre-inches*
11.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 05/29/2023: Corn, silage

Field name: 12 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.65	0.00	0.00	445.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.19	40.80	97.56	4,848.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	167.40	39.06	195.30	1,106.18
Nutrient balance	61.79	1.74	-97.74	3,742.26
Applied to removed ratio	1.37	1.04	0.50	4.38

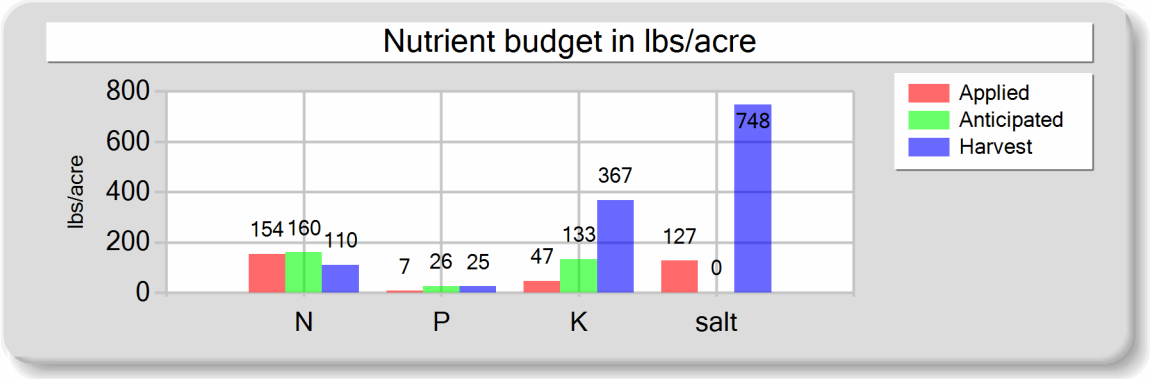
Fresh water applied
8,208,373.00 *gallons*
302.29 *acre-inches*
37.79 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

13 - 10/04/2022: Oats, silage-soft dough

Field name: 13 Crop: Oats, silage-soft dough Plant date: 10/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.46	0.00	0.00	126.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.74	6.90	46.94	126.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	109.60	24.66	367.16	747.59
Nutrient balance	44.14	-17.76	-320.22	-620.79
Applied to removed ratio	1.40	0.28	0.13	0.17

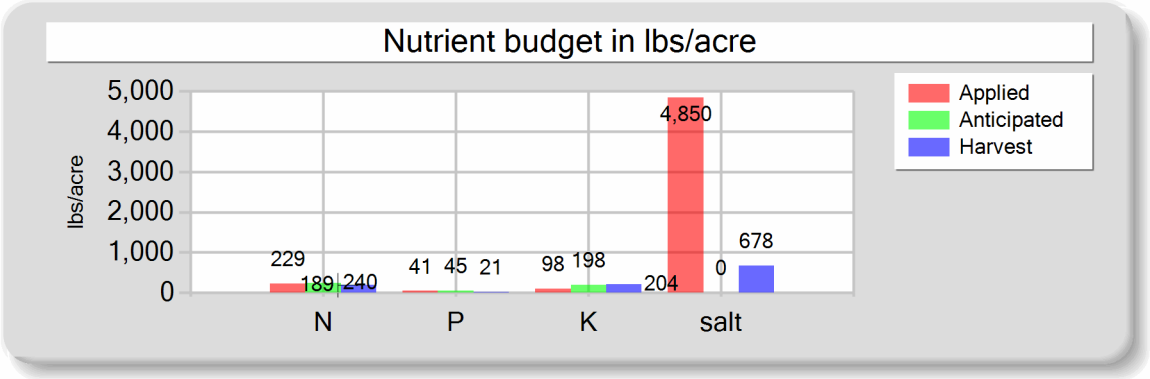
Fresh water applied	
7,889,733.00	*gallons*
290.55	*acre-inches*
10.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 06/19/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.68	0.00	0.00	446.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.22	40.80	97.56	4,850.13
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	188.64	20.96	204.36	677.85
Nutrient balance	40.58	19.84	-106.80	4,172.28
Applied to removed ratio	1.22	1.95	0.48	7.16

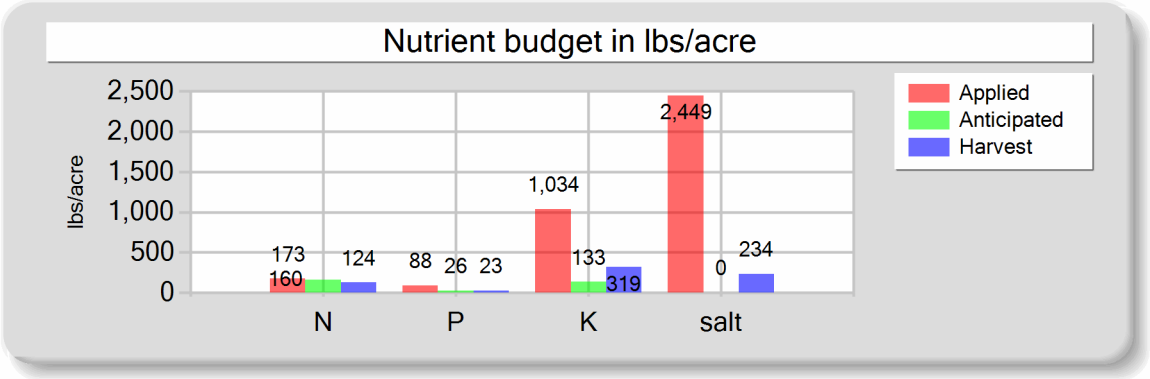
Fresh water applied	
27,808,245.00	*gallons*
1,024.08	*acre-inches*
37.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/08/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.54	88.48	1,034.28	2,321.78
Fresh water	2.48	0.00	0.00	127.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.02	88.48	1,034.28	2,449.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	124.08	22.56	318.66	233.85
Nutrient balance	48.94	65.92	715.62	2,215.60
Applied to removed ratio	1.39	3.92	3.25	10.47

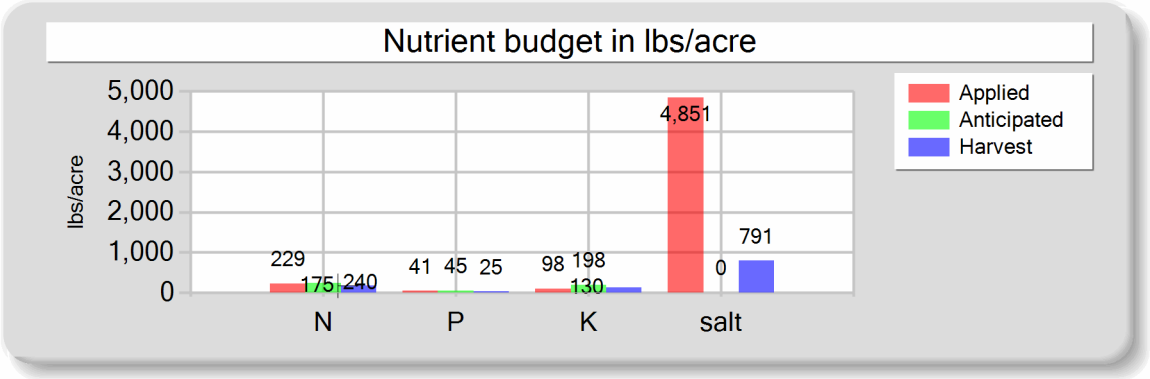
Fresh water applied	
4,707,144.00	*gallons*
173.35	*acre-inches*
10.83	*inches/acre*

Process wastewater applied	
1,412,105.00	*gallons*
52.00	*acre-inches*
3.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/13/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.69	0.00	0.00	447.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.23	40.80	97.56	4,850.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	175.00	25.00	130.00	791.20
Nutrient balance	54.23	15.80	-32.44	4,059.32
Applied to removed ratio	1.31	1.63	0.75	6.13

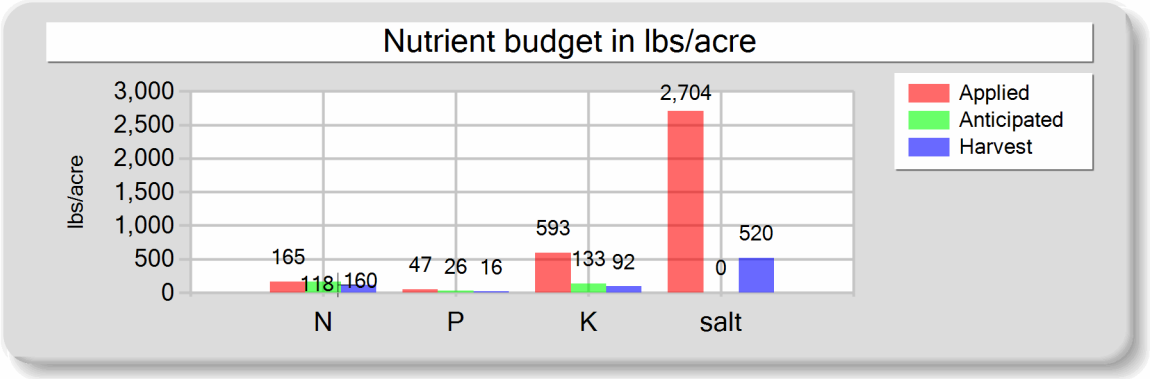
Fresh water applied	
16,493,379.00	*gallons*
607.40	*acre-inches*
37.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/01/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	156.71	47.07	592.87	2,631.39
Fresh water	1.40	0.00	0.00	72.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.11	47.07	592.87	2,703.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	117.90	15.72	91.70	520.34
Nutrient balance	47.21	31.35	501.17	2,183.18
Applied to removed ratio	1.40	2.99	6.47	5.20

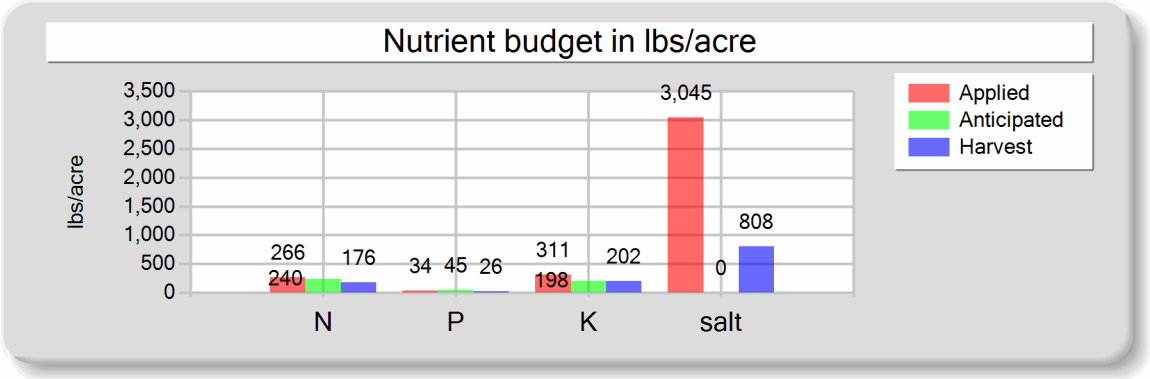
Fresh water applied	
4,987,080.00	*gallons*
183.66	*acre-inches*
6.12	*inches/acre*

Process wastewater applied	
2,083,060.00	*gallons*
76.71	*acre-inches*
2.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/14/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	156.77	19.31	295.69	2,600.08
Fresh water	8.65	0.00	0.00	445.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.42	34.31	310.69	3,045.31
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	176.12	25.90	202.02	808.39
Nutrient balance	90.30	8.41	108.67	2,236.92
Applied to removed ratio	1.51	1.32	1.54	3.77

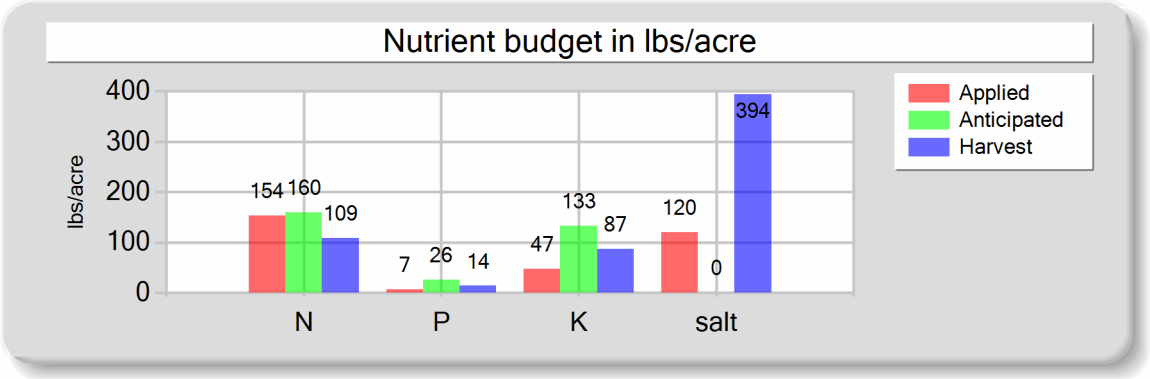
Fresh water applied	
30,780,540.00	*gallons*
1,133.54	*acre-inches*
37.78	*inches/acre*

Process wastewater applied	
2,290,977.00	*gallons*
84.37	*acre-inches*
2.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/01/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.34	0.00	0.00	120.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.61	6.90	46.94	120.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	108.80	13.60	87.04	394.39
Nutrient balance	44.81	-6.70	-40.10	-274.13
Applied to removed ratio	1.41	0.51	0.54	0.30

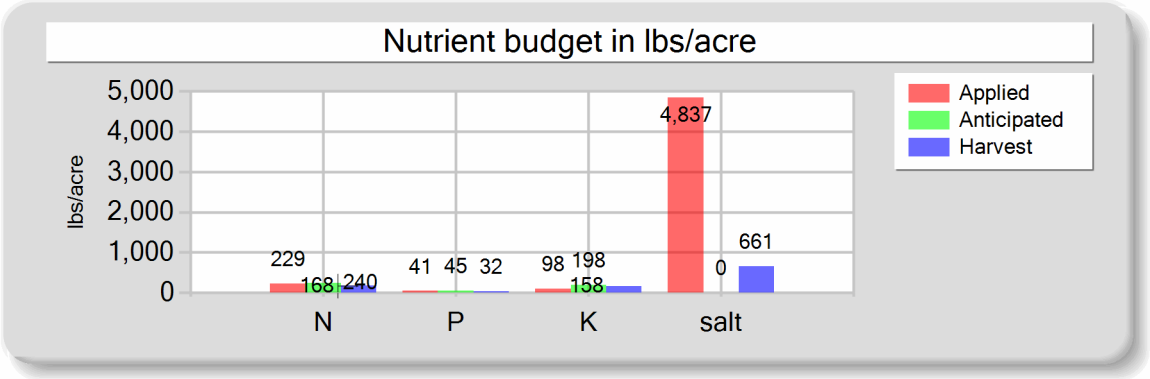
Fresh water applied	
10,253,826.00	*gallons*
377.61	*acre-inches*
10.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/01/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.43	0.00	0.00	433.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.97	40.80	97.56	4,837.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	168.32	31.56	157.80	660.66
Nutrient balance	60.65	9.24	-60.24	4,176.44
Applied to removed ratio	1.36	1.29	0.62	7.32

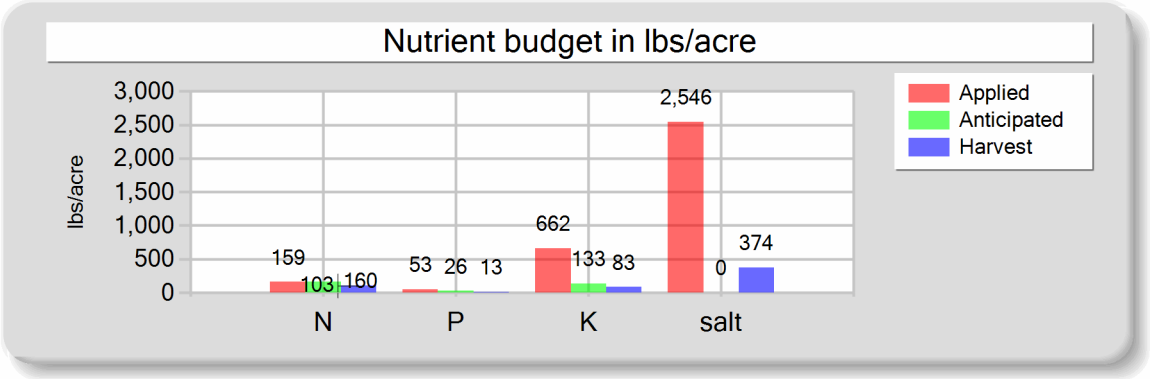
Fresh water applied	
36,996,063.00	*gallons*
1,362.44	*acre-inches*
36.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/02/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.71	52.96	662.01	2,475.37
Fresh water	1.36	0.00	0.00	70.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.07	52.96	662.01	2,545.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	103.20	12.90	82.56	374.09
Nutrient balance	55.87	40.06	579.45	2,171.43
Applied to removed ratio	1.54	4.11	8.02	6.80

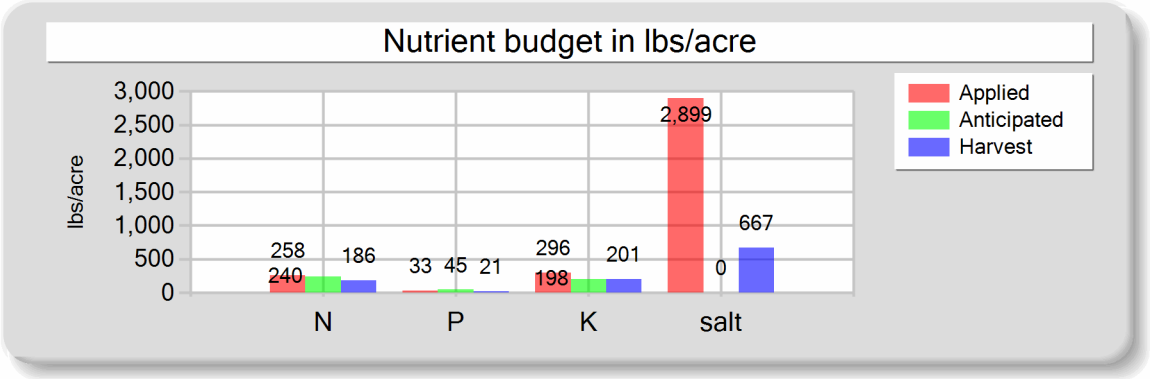
Fresh water applied	
4,041,130.00	*gallons*
148.82	*acre-inches*
5.95	*inches/acre*

Process wastewater applied	
1,740,368.00	*gallons*
64.09	*acre-inches*
2.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/14/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	149.22	18.38	281.45	2,474.82
Fresh water	8.25	0.00	0.00	424.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.47	33.38	296.45	2,899.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	185.76	20.64	201.24	667.50
Nutrient balance	72.71	12.74	95.21	2,231.99
Applied to removed ratio	1.39	1.62	1.47	4.34

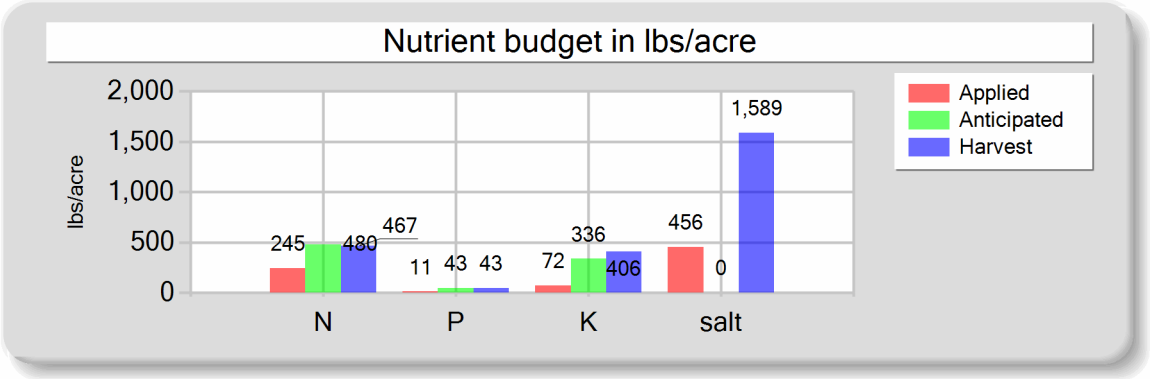
Fresh water applied	
24,466,064.00	*gallons*
901.00	*acre-inches*
36.04	*inches/acre*

Process wastewater applied	
1,817,175.00	*gallons*
66.92	*acre-inches*
2.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 09/30/2022: Alfalfa, hay

Field name: 6 Crop: Alfalfa, hay Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	221.96	10.62	72.22	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.85	0.00	0.00	455.90
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	244.81	10.62	72.22	455.90
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	466.56	43.44	406.11	1,589.36
Nutrient balance	-221.74	-32.82	-333.90	-1,133.47
Applied to removed ratio	0.52	0.24	0.18	0.29

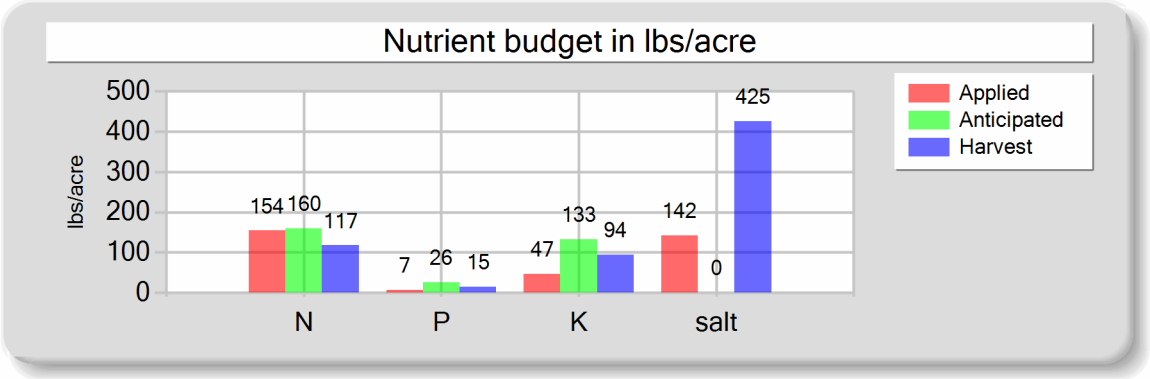
Fresh water applied	
37,821,529.00	*gallons*
1,392.84	*acre-inches*
38.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 10/03/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.76	0.00	0.00	142.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.03	6.90	46.94	142.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	117.35	14.67	93.88	425.40
Nutrient balance	36.68	-7.77	-46.94	-283.40
Applied to removed ratio	1.31	0.47	0.50	0.33

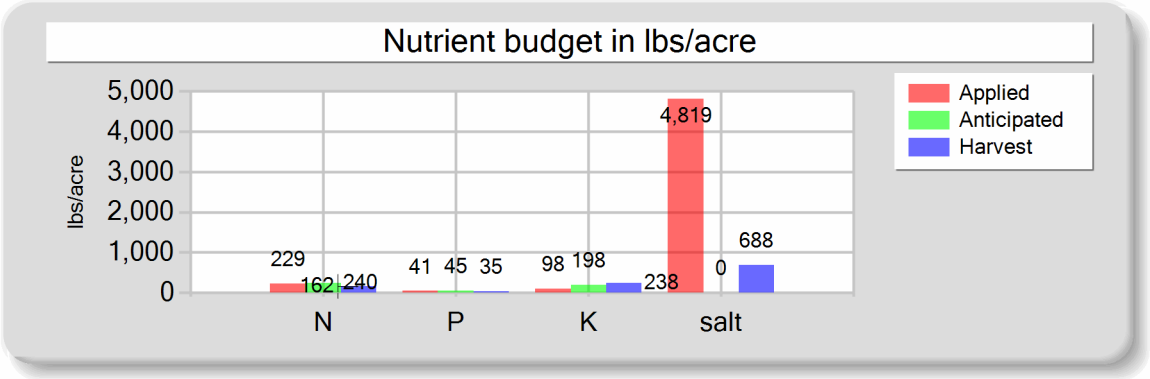
Fresh water applied	
8,507,964.00	*gallons*
313.32	*acre-inches*
12.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/30/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.08	0.00	0.00	415.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.62	40.80	97.56	4,819.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	161.92	35.42	237.82	687.65
Nutrient balance	66.70	5.38	-140.26	4,131.38
Applied to removed ratio	1.41	1.15	0.41	7.01

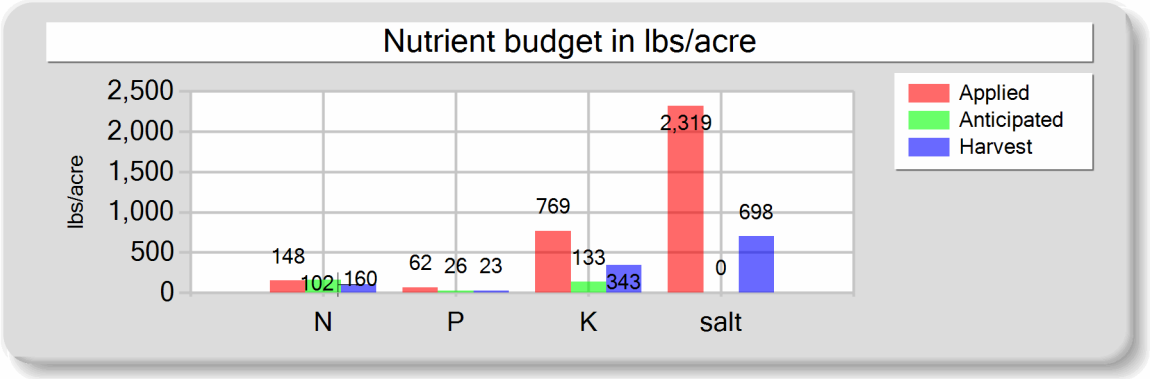
Fresh water applied	
24,914,934.00	*gallons*
917.53	*acre-inches*
35.29	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/03/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	138.78	62.11	768.85	2,183.40
Fresh water	2.63	0.00	0.00	135.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	148.41	62.11	768.85	2,318.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	102.40	23.04	343.04	698.48
Nutrient balance	46.01	39.07	425.81	1,620.19
Applied to removed ratio	1.45	2.70	2.24	3.32

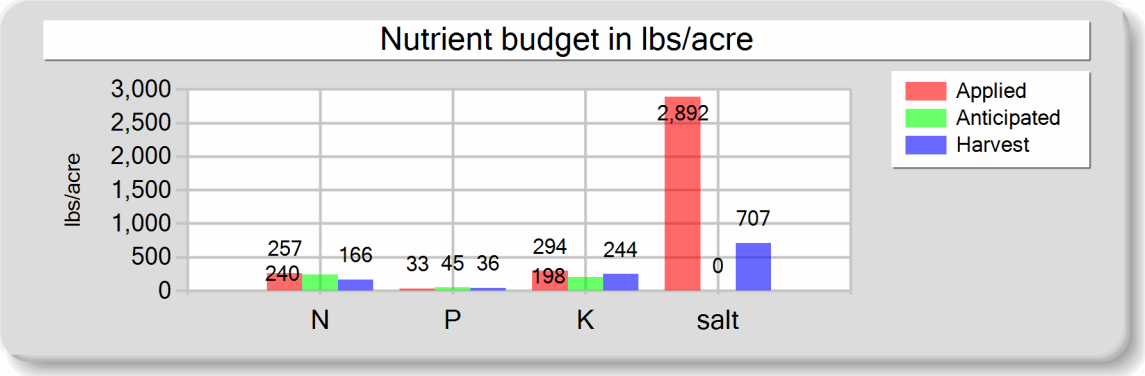
Fresh water applied	
5,922,892.00	*gallons*
218.12	*acre-inches*
11.48	*inches/acre*

Process wastewater applied	
1,311,405.00	*gallons*
48.29	*acre-inches*
2.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/29/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	147.74	18.20	278.67	2,450.35
Fresh water	8.58	0.00	0.00	441.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.32	33.20	293.67	2,892.17
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.40	36.40	244.40	706.68
Nutrient balance	90.92	-3.20	49.27	2,185.49
Applied to removed ratio	1.55	0.91	1.20	4.09

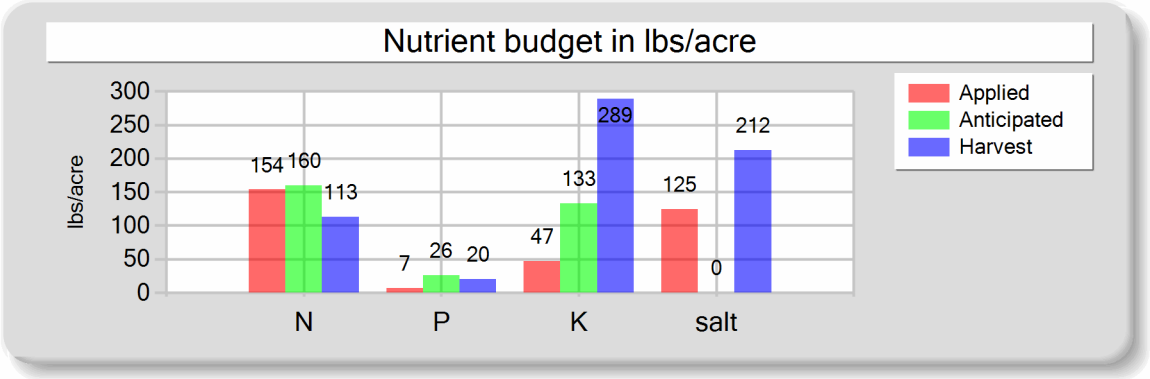
Fresh water applied	
19,344,938.00	*gallons*
712.41	*acre-inches*
37.50	*inches/acre*

Process wastewater applied	
1,367,400.00	*gallons*
50.36	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

P3 - 10/04/2022: Oats, silage-soft dough

Field name: P3 Crop: Oats, silage-soft dough Plant date: 10/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.27	6.90	46.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.42	0.00	0.00	124.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.70	6.90	46.94	124.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.61	20.47	289.19	212.23
Nutrient balance	41.09	-13.57	-242.25	-87.46
Applied to removed ratio	1.36	0.34	0.16	0.59

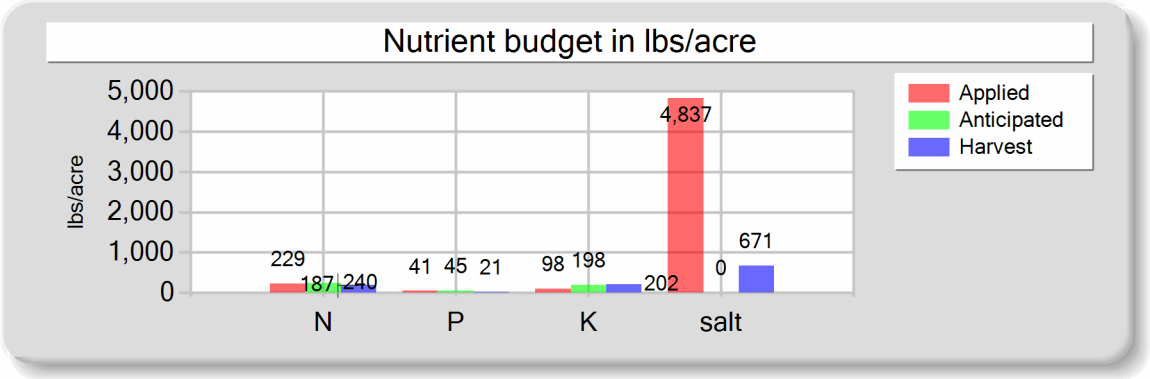
Fresh water applied	
3,737,873.00	*gallons*
137.65	*acre-inches*
10.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

P3 - 06/19/2023: Corn, silage

Field name: P3 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	94.00	15.00	15.00	0.00
Dry manure	119.54	25.80	82.56	4,403.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	433.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.96	40.80	97.56	4,836.65
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	186.87	20.76	202.44	671.48
Nutrient balance	42.09	20.04	-104.88	4,165.18
Applied to removed ratio	1.23	1.97	0.48	7.20

Fresh water applied	
12,985,459.00	*gallons*
478.21	*acre-inches*
36.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Solid Manure

Sample and source description: 2022 Fall Solid Manure

Sample date: 11/09/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	1,000.00	6,800.00							
DL	100.00	100.00	100.00							

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/12/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,000.00	9,600.00							51.20
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,800.00	13,100.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

WWSQ3

Sample and source description: WWSQ3

Sample date: 07/06/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	246.00	151.00	0.00	0.00	35.30	300.00								3,940.00	3,440
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 11/09/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	316.00	227.00	0.00	0.00	40.20	542.00								4,690.00	5,700
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	213.00	140.00	0.00	0.00	133.00	1,630.00								3,140.00	3,080
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 05/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	188.00	0.00	0.00	62.00	384.00								4,190.00	3,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	246.00	162.00	0.00	0.00	30.30	464.00	105.00	116.00	214.00	2,380.00	0.00	344.00	198.00	5,090.00	4,080
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	249.00	146.00	0.00	0.00	43.80	321.00								2,980.00	5,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

OID Canal

Sample description: OID Canal

Sample date: 08/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,000.00	14,000.00		8.71
DL	100.00	100.00	100.00		0.01

1 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	700.00	2,800.00		4.00
DL	100.00	100.00	100.00		0.01

10 - 10/07/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	900.00	13,400.00		8.58
DL	100.00	100.00	100.00		0.01

10 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	500.00	2,300.00		4.40
DL	100.00	100.00	100.00		0.01

11 - 10/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,400.00	400.00	3,600.00		6.74
DL	100.00	100.00	100.00		0.01

11 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	400.00	2,400.00		2.90
DL	100.00	100.00	100.00		0.01

12 - 10/05/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,400.00	400.00	4,700.00		9.86
DL	100.00	100.00	100.00		0.01

12 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	700.00	3,500.00		5.60
DL	100.00	100.00	100.00		0.01

13 - 10/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,000.00	14,000.00		8.71
DL	100.00	100.00	100.00		0.01

13 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	400.00	3,900.00		4.20
DL	100.00	100.00	100.00		0.01

2 - 10/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	800.00	11,300.00		2.71
DL	100.00	100.00	100.00		0.01

2 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	500.00	2,600.00		4.30
DL	100.00	100.00	100.00		0.01

3 - 10/01/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	600.00	3,500.00		5.74
DL	100.00	100.00	100.00		0.01

3 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	500.00	3,900.00		5.10
DL	100.00	100.00	100.00		0.01

4 - 10/01/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	600.00	3,800.00		5.73
DL	100.00	100.00	100.00		0.01

4 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	600.00	3,000.00		4.00
DL	100.00	100.00	100.00		0.01

5 - 10/02/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	500.00	3,200.00		5.62
DL	100.00	100.00	100.00		0.01

5 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	500.00	3,200.00		6.80
DL	100.00	100.00	100.00		0.01

6 - 09/30/2022: Alfalfa, hay

Alfalfa Hay

Sample and source description: Alfalfa Hay

Sample date: 12/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	2,300.00	21,500.00		9.54
DL	100.00	100.00	100.00		0.01

7 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,800.00	600.00	5,600.00		8.12
DL	100.00	100.00	100.00		0.01

7 - 05/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,700.00	500.00	2,600.00		3.70
DL	100.00	100.00	100.00		0.01

9 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	700.00	6,500.00		9.41
DL	100.00	100.00	100.00		0.01

9 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	700.00	4,700.00		4.50
DL	100.00	100.00	100.00		0.01

P3 - 10/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	900.00	13,400.00		8.58
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

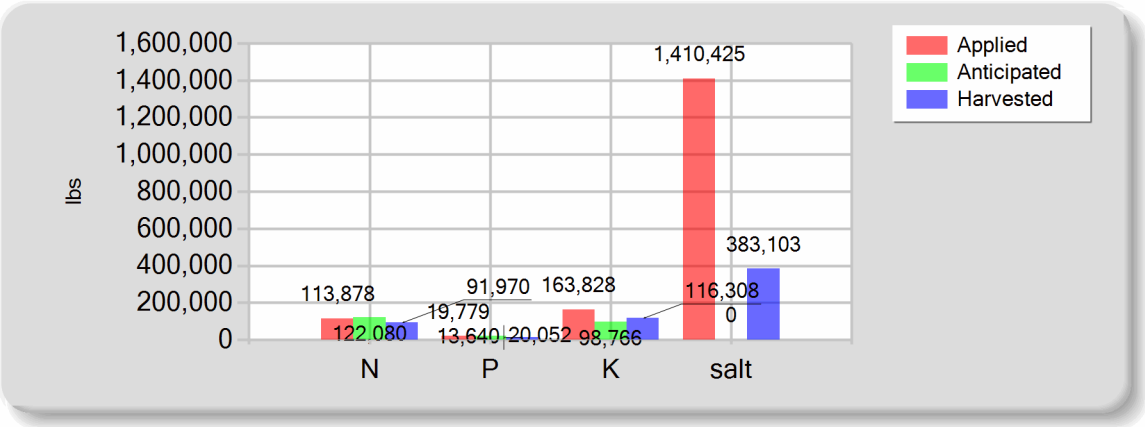
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

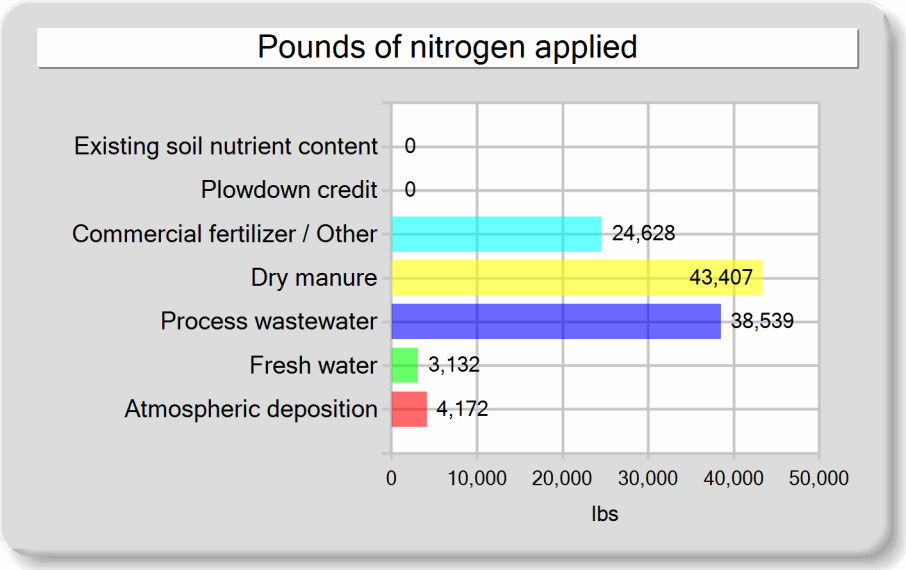
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

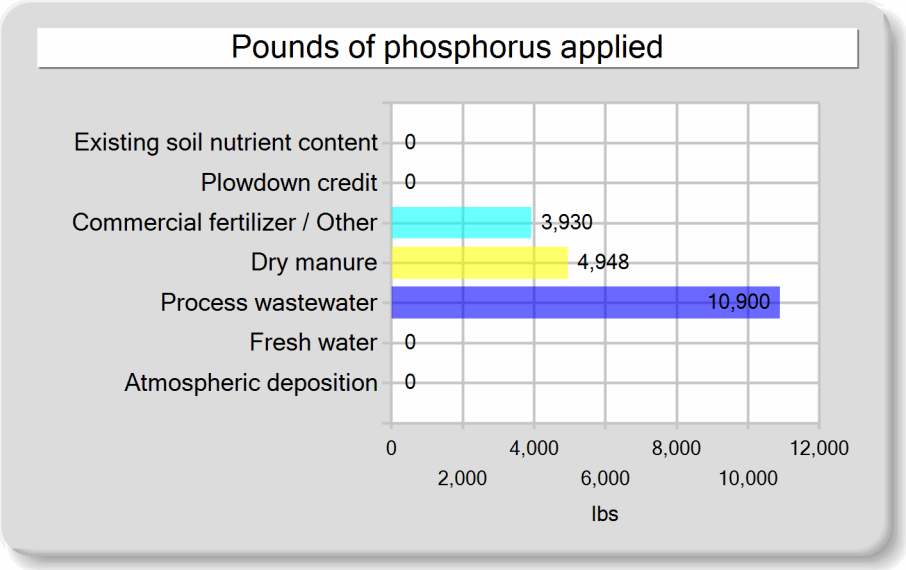
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	24,628.00	3,930.00	3,930.00	0.00
Dry manure	43,407.34	4,948.40	20,367.29	629,657.60
Process wastewater	38,538.81	10,900.12	139,530.45	619,510.60
Fresh water	3,132.10	0.00	0.00	161,256.52
Atmospheric deposition	4,172.00	0.00	0.00	0.00
Total nutrients applied	113,878.25	19,778.52	163,827.74	1,410,424.73
Anticipated crop nutrient removal	122,080.00	20,052.40	98,765.60	0.00
Actual crop nutrient removal	91,970.06	13,639.92	116,308.25	383,103.16
Nutrient balance	21,908.19	6,138.60	47,519.49	1,027,321.56
Applied to removed ratio	1.24	1.45	1.41	3.68

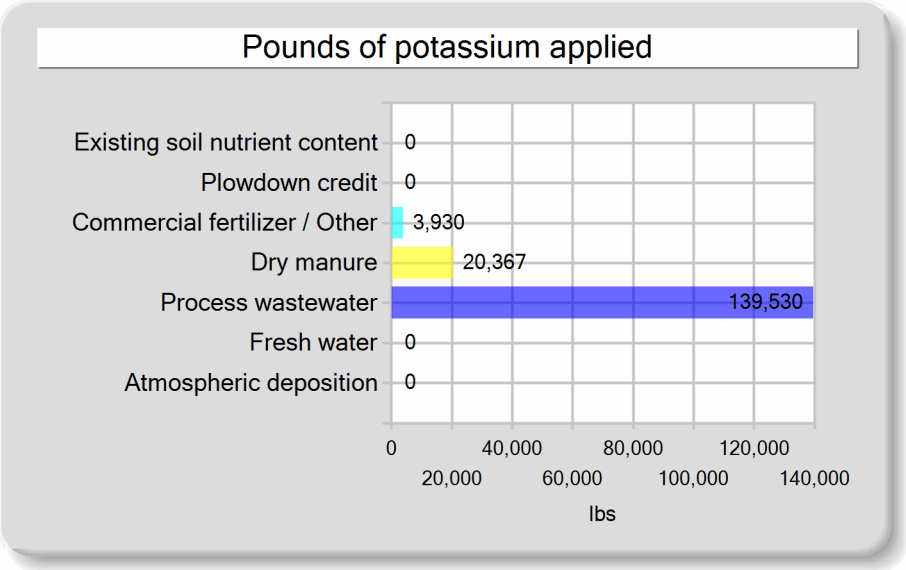
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

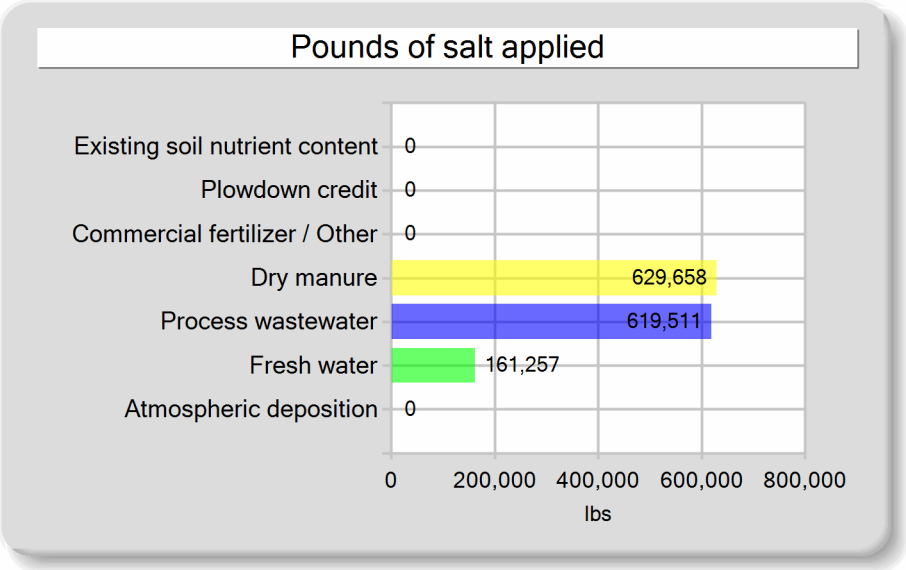


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

A combination of lower than anticipated yields (12.8-13.4 tons per acre harvested vs 16 tons per acre anticipated) and lower than desired nitrogen removal (6.51-8.42 lbs of nitrogen removed vs 10 lbs of nitrogen anticipated) resulted in fields 1, 5, 9, and 10 for winter forage to have higher than desired field ratios.

Corn Silage for Fields 3 (1.51), 9 (1.55) and 10 (1.53) had lower than desired nitrogen removal rates (6.4-6.81 lbs of nitrogen removed vs 8 lbs of nitrogen anticipated) caused these fields to be above the 1.4 threshold.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

DeeAnn Schnibbe (May 20, 2024 11:11 PDT)	kellye caton (May 20, 2024 09:34 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Barbara Beckett Bypass Trust	Cody Caton
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 20, 2024	May 20, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
DJ Caton Dairy
15206 Steinegual Rd
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/20/23 11:15 AM
SUBMITTED BY: Courier
ANALYZED DATE: 12/28/23

Process Waste Water (H3)
SOURCE: Lagoon Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units	Lbs/10000 Gal as Rec'd
WWQT 4	T3354007A	Total Dissolved Solids (TDS)	SM 2540 C-1997	5140	mg/L	429
		Total Phosphorus (TP)	EPA 200.7	43.8	mg/L	3.65
		Electrical Conductivity (EC)	EPA 120.1 and	2.98	mmhos/cm	N/A
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-	146	mg/L	12.1
		Total Kjeldahl Nitrogen (TKN)	SM 4500-NH3 C-	249	mg/L	20.8
		Total Potassium (TK)	EPA 200.7	321	mg/L	26.8
		Soluble Salts (SALT-SOL)	Calculation	1910	mg/L	159

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			146 (T3354007A)	146 (null)	0.000
Total Dissolved Solids (TDS)	101			10400 (T3354037A)	10800 (Dup)	3.95
Total Kjeldahl Nitrogen (TKN)	103			249 (T3354007A)	241 (null)	3.43
Total Phosphorus (TP)	98.8	100 (T3353011A)	100 (T3353011A)			0.020
Total Potassium (TK)	101	100 (T3353011A)	103 (T3353011A)			2.63

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

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Compliance Analysis Report

Customer
DJ Caton Dairy
15206 Steinegual Rd
Escalon, CA 95320

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/20/23 11:15 AM
SUBMITTED BY: Courier
ANALYZED DATE: 12/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well	T3354007B	Electrical Conductivity (EC)	EPA 120.1 and	0.552	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	14.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	353	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	104	12.7 (T3354012A)	13.9 (T3354012A)	8.73

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: F+R AG SERVICE

Client: CATON FARMS
Mailing address: 15206 STEINEGUL RD
(not facility address) ESCALON CA 95320
Phone#: 209 456 3266

Sampled By: CODY CATON
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/20	10 AM	WW QT 4		18.6°C			X			
2	12/20	10 AM	DOMESTIC WELL		19.0°C		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Cody Caton	CATON FARMS	12/20 11:15 AM			
2nd		DENELE ANALYTICAL, INC.	12-20 11:15			T3354007

Required to be submitted to any agency? (YES / NO)

2023 Annual Report- Caton Farms

Final Audit Report 2024-05-20

Created:	2024-05-20
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAKXskfByj2NHNFhTTwI9ZzAJAUMPG4OI6

"2023 Annual Report- Caton Farms" History

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Signer dschnibbe@gmail.com entered name at signing as DeeAnn Schnibbe
2024-05-20 - 6:11:05 PM GMT

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2024-05-20 - 6:11:07 PM GMT